Demonstration of Proposed Features

Date	30 June 2026
Team ID	SWTID-2026-4836
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	1 Mark

Demonstration of Proposed Features:

This document captures all the features that were proposed during the project planning phase and tracks whether each feature was successfully implemented and demonstrated. It serves as evidence of the team's ability to deliver on the proposed solution.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Crop Recommendation Prediction	Predicts suitable crops based on N, P, K, temperature, humidity, pH, and rainfall values.	Implemented	Yes	Model generates crop recommendations successfully.
2	Data Analysis and Visualization	Performs univariate, bivariate, and multivariate analysis on agricultural data.	Implemented	Yes	Graphs and charts generated successfully.
3	Machine Learning Model Training	Trains Logistic Regression and K-Means models for crop prediction.	Implemented	Yes	Models trained with good accuracy.
4	Interactive Web Interface	Provides Home, About, and FindYourCrop pages for user interaction.	Implemented	Yes	User-friendly interface developed using HTML and Flask.
5	Backend Integration	Connects the Flask application with the trained machine learning model.	Implemented	Yes	Prediction requests are processed successfully.
6	Season-Based Crop Analysis	Provides recommendations based on seasonal environmental conditions.	Partial	No	Enhancement can be added in future versions.

Feature Implementation Summary:

Total Features Proposed	6
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	83.33%